

DIYATH WICKRAMARATNE

SENIOR FULL-STACK SOFTWARE ENGINEER | NEXT.JS, REACT, NODE.JS, AWS SERVERLESS & AI ENGINEERING

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PROFESSIONAL SUMMARY

Senior Full-Stack Software Engineer with 10+ years of experience building scalable production applications using React, Next.js, TypeScript, Node.js, AWS serverless technologies, REST APIs, microservices, and Infrastructure as Code with AWS SAM and CDK. Experienced in multi-tenant SaaS systems, event-driven architecture, automated testing, CI/CD, observability, and production support. Hands-on experience using Claude Code and agentic engineering workflows across the SDLC, with human review for architecture, security, correctness, testing, and maintainability.

CORE TECHNOLOGIES

Languages: TypeScript, JavaScript, C#, Python, SQL, PHP

Frontend: Next.js, React, TypeScript, Redux, Material UI, Angular, Vue.js

Backend & Architecture: Node.js, .NET Core, REST APIs, Microservices, Multi-Tenant SaaS Architecture, Event-Driven Systems, Asynchronous Workflows, Real-Time Applications

AI & Agentic Engineering: Claude Code, CLAUDE.md, Agent Skills, MCP, Agents, Subagents, Controlled Tool Integrations, Context Engineering, Human-in-the-Loop Review

AWS & Cloud: AWS Lambda, DynamoDB, AWS SAM, AWS CDK, EC2, S3, CloudFront, CloudWatch, CodePipeline, Docker, CI/CD

Data & Integration: MongoDB, Mongoose, MySQL, SQL Server, Webhooks, Jira, Bitbucket

Testing & Observability: Jest, Playwright, Sentry, Codacy, ESLint, Code Review, Production Support

PROFESSIONAL EXPERIENCE

Senior Full Stack Software Engineer | Optergy | Australia | October 2024 – July 2026

- Used Claude Code throughout the SDLC, including requirements analysis, solution planning, implementation, debugging, automated testing, code review, documentation, multi-file change review, architecture validation, security review, and human-in-the-loop approval.
- Built controlled LLM tool integrations that enabled CRUD access to application data through predefined tools, permissions, validation rules, and reviewable execution paths.
- Developed responsive React, TypeScript, and Material UI interfaces and scalable Node.js APIs supporting complex application workflows and operational data.
- Developed and maintained scalable full-stack capabilities within a multi-tenant SaaS architecture using React, TypeScript, Node.js, MongoDB, and AWS.
- Built and maintained project-specific CLAUDE.md context and reusable Agent Skills covering application architecture, repository structure, coding standards, testing strategies, security considerations, and repeatable engineering workflows.
- Integrated Claude Code with Jira through MCP, enabling agent-assisted development from ticket requirements and acceptance criteria while reviewing assumptions, architecture, security, code quality, testing approach, and documentation changes.
- Guided Claude Code agents, subagents, and agent teams across full-stack feature delivery, backend API implementation, multi-file change review, test creation, debugging, technical impact analysis, and documentation while owning final implementation decisions.
- Integrated automated verification into AI-assisted engineering workflows using Jest unit and integration tests, Playwright UI/E2E testing, Sentry monitoring, Codacy, ESLint, and human review checkpoints.
- Designed and optimized MongoDB schemas using Mongoose, improving query performance through indexing and schema refinement.
- Managed AWS infrastructure, Bitbucket Pipelines, CI/CD workflows, and Sentry monitoring across build, test, deployment, and production support activities, supporting release reliability, observability, performance, security, and incident response.

Senior Full Stack Developer | Step Global | Australia | October 2022 – September 2024

- Delivered core ASGARD platform capabilities using Next.js, React, Node.js, TypeScript, .NET Core, MongoDB, and AWS, owning full-stack features across frontend, backend APIs, deployment, and production workflows.

- Migrated legacy Node.js/JavaScript services to TypeScript while preserving existing functionality and improving maintainability.
- Implemented AWS Secrets Manager and JWT authentication, strengthening API security and credential management.
- Configured AWS CodePipeline and delivered frontend assets through Amazon S3 and CloudFront for automated and scalable deployments.
- Built backend services integrating Hikvision Gateway and NVR systems to support event-linked video workflows, centralized AWS S3 uploads, retrieval, and processing.
- Designed event-driven backend services processing real-time GPS, telematics, vehicle-event, speed, and video workflows across 300+ connected trucks, integrating APIs, MongoDB GeoJSON, HERE Maps, Hikvision/NVR systems, and AWS S3.
- Developed asynchronous telemetry processing workflows for truck GPS location, vehicle events, speed events, video events, logging, error investigation, and operational tracking to support fleet monitoring and analysis.
- Delivered event-linked video playback, live streaming, and configurable Wi-Fi/cellular upload workflows for real-time fleet monitoring.
- Built the Smart eDriver regulatory-compliance frontend using React, Formik, and Redux for Australian transport operations.

Senior Software Engineer | Orivet Pet Solutions | Australia | January 2017 – September 2022

- Helped scale the production platform from approximately 1,000 to more than 162,000 users through backend architecture improvements, AWS cloud migration, production modernization, and API-driven system design.
- Developed scalable Node.js and .NET Core APIs and microservices as the platform transitioned toward service-oriented, event-driven, and serverless architecture.
- Migrated workloads to AWS serverless architecture, implementing AWS Lambda-based microservices, REST APIs, CI/CD pipelines, and Dockerized services to support scalable production operations.
- Used AWS SAM and AWS CDK to define and deploy serverless infrastructure as part of the platform's AWS modernization, supporting repeatable deployment of Lambda-based services and related cloud resources.
- Maintained the legacy PHP/Laravel production application and administration panel where needed for historical platform continuity while modernization progressed.
- Delivered custom analytics dashboards using Highcharts and D3 for veterinary genetic and health reporting.
- Built a Dockerized .NET Core PDF-generation API for automated report creation.
- Integrated third-party services including Google APIs and Shopify webhooks, supporting API reliability, asynchronous workflows, and external platform connectivity.

Software Engineer | Eutech Cybernetics Lanka (Pvt) Ltd. | Sri Lanka | June 2015 – December 2016

- Delivered projects for AVAYA, LTA, KTPH, and BHP Marketing Asia Pte Ltd., Singapore.
- Built smart-workplace and facility-management applications using C#, Python, JavaScript, and Highcharts.
- Developed backend integrations connecting access-card systems, electronic displays, application services, and Philips Hue controls using C# and Python.
- Developed custom Permit-to-Work applications for the Iviva cloud using JavaScript and jQuery.
- Customized Synergy FM with a corrective work-order management system.

Associate Software Engineer | Victory Information (Pvt) Ltd. | Sri Lanka | February 2014 – June 2015

- Maintained enterprise ERP software systems and provided customer support and technical assistance.
- Developed and maintained Victory POS using C#, WPF/WCF, SQL Server, and Crystal Reports.
- Designed and customized Victory ERP modules by gathering user requirements and implementing client-specific business workflows and modifications.

EDUCATION

Bachelor of Science in Computer Science

University College Dublin, Ireland